

KSHITIJ DESAI

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SUMMARY

AI Engineer with 3 years of experience building and deploying production AI/ML systems. Founding engineer at Calld.AI delivering enterprise Voice AI infrastructure with sub-second latency at scale. Deep experience across real-time inference serving, LLM fine-tuning, RAG, and agentic systems.

TECHNICAL SKILLS

Languages: Python, C++, SQL

ML / Deep Learning: PyTorch, JAX, Scikit-Learn, XGBoost, Reinforcement Learning, NLP, Computer Vision

LLM & Agentic: Fine-tuning (LoRA/QLoRA), RLHF, RAG, LangGraph, LangChain, MCP

Inference & Serving: vLLM, FastAPI, LiveKit, TTS/STT deployment, real-time streaming pipelines

Data & Infra: PostgreSQL, MongoDB, Redis, Milvus, Qdrant, PySpark, AWS, Azure, Databricks, CI/CD, GPU autoscaling

EXPERIENCE

AI Engineer (Founding Engineer)

Calld.AI | Part-time

Mar 2025 – Present

Melbourne, Australia

- One of 5 founding engineers building enterprise Voice AI for call centre automation, serving live production customer calls.
- Deployed real-time TTS/STT inference pipelines sustaining 10+ concurrent calls per instance at **<1s end-to-end latency**.
- Designed autoscaled CPU/GPU orchestration with load balancing, cutting cold-start overhead and maintaining **99%+ uptime**.
- Built end-to-end observability (Grafana dashboards, alerting) enabling production incident detection and resolution.

AI Engineer

ExpertEase.AI | Contract

Dec 2024 – Mar 2025

Adelaide, Australia

- Led Voice AI development (team of 4) for real-time conversational systems handling 10+ concurrent sessions per instance.
- Built multi-modal RAG chatbot ingesting 5+ document types (CSV, Excel, PDF, web) for 3 enterprise clients, delivering source-attributed answers via vector retrieval.

Specialist Data Scientist

SOLV (Standard Chartered Research & Technology)

Jul 2022 – May 2024

Bengaluru, India

- Built an internal LLM-based Document Q&A retrieval system and integrated an image captioning service into the platform.
- Improved buyer segmentation across **500K+ records** using clustering and cohort modelling; insights adopted by 6 stakeholder teams including CEO Office and Risk & Compliance.
- Developed end-to-end ML pipeline for delivery-time and return-order prediction achieving **80%+ accuracy**.
- Mentored 5 junior data scientists and automated data ingestion workflows saving **10 hrs/week**.

Programmer Analyst Trainee

Cognizant Technology Solutions

Oct 2021 – Jul 2022

Chennai, India

EDUCATION

Master of Artificial Intelligence & Machine Learning

The University of Adelaide **GPA: 6.91 / 7.00**

May 2024 – May 2026

Adelaide, SA, Australia

- Capstone: *Investigation of Adaptivity within SMC in Reinforcement Learning* – Sequential Monte Carlo methods for RL policy optimisation.

PROJECTS

Self-Hosted WebRTC Video Calling Platform | *LiveKit, WebRTC, VPS, VPN Tunnel*

- Self-hosted a LiveKit server exposed via VPS and VPN tunnel, serving multi-participant video calls live at kshitijdesai.com/apps/talk.

LLM Fine-Tuning for Low-Resource Translation | *Qwen, LoRA, QLoRA, PEFT, chrF2*

- Benchmarked English → Kannada PEFT fine-tuning on 16.7K/4.0K train/val pairs; Qwen3.5-4B + LoRA reached **60.20 chrF2**, beating a zero-shot 9B model by **+8.75**.

Document Search Agent | *BM25, RAG, ZeroEntropy, LangSmith, Python*

- Built a two-stage PDF RAG agent over 3,241 chunks from 37 PDFs (BM25 top-100 retrieval, reranking, context expansion, source citations), passing the full easy/hard QA eval suite.

LLM Code Execution Agent | *Docker, LangGraph, LangSmith, Python*

- Built a LangGraph agent routing compute, visualisation, and chat tasks to Docker-backed sub-agents with LangSmith tracing and parallel eval diagnostics across OpenAI/Gemini eval sets.

Kaggle Hackathons (IISc) | *PyTorch, CNNs, Transfer Learning, Audio DSP*

- Ranked **2nd** in Caltech-256 image classification (257 classes, **0.945** accuracy) and **2nd** in musical audio emotion classification (**0.804** accuracy) using MFCC/Mel spectrogram features.

CERTIFICATIONS

Deep Learning Foundations & Applications | *Indian Institute of Science*

Oct 2022 – Nov 2023

Data Science Immersive Bootcamp | *Analytics Vidhya*

Oct 2021 – Mar 2022